Bogyeom Park

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RESEARCH INTEREST

I design and study **human-centered agentic AI systems** that support learning, decision-making, and accessibility, at the intersection of Human-AI Interaction, AI in Education, and Learning Analytics.

- **Agentic AI for learning and decision support -** designing agents that reason with people, use tools, and adapt support while preserving human goals and oversight
- **Human-centered interaction and evaluation -** analyzing conversational and behavioral traces to understand engagement, accessibility, and real-world outcomes

EDUCATION

Seoul National University of Science and Technology — Seoul, South Korea
Ph.D. in Applied Artificial Intelligence — *Mar. 2023 - Present*

- Advisor: Kyoungwon Seo

Seoul National University of Science and Technology — Seoul, South Korea
B.S. in Electrical and Information Engineering — *Mar. 2019 - Feb. 2023*

- Thesis: Online Learning Support System Based on Facial Recognition

HONORS & AWARDS

- AI SeoulTech Graduate Scholarship (KRW 10,000,000), Seoul Scholarship Foundation, 2025
- Best Paper Award (co-author), HCI Korea 2025
- Best Student Paper Bronze Award, IEEE Seoul Section, 2024
- National Science and Engineering Undergraduate Scholarship (full tuition), Korea Student Aid Foundation, 2021-2022
- Best Capstone Design Award, Seoul National University of Science and Technology, 2022

JOURNAL ARTICLES

Integrating Biomarkers From Virtual Reality and Magnetic Resonance Imaging for the Early Detection of Mild Cognitive Impairment Using a Multimodal Learning Approach: Validation Study
Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo
Journal of Medical Internet Research, 26, e54538 (2024) - JCR Q1; Top 3%; JIF 8.2 (2025 JCR)

INTERNATIONAL CONFERENCE PAPERS

Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot
Bogyeom Park and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

How Self-Disclosing Chatbots Influence Student Engagement, Assessment Accuracy, and Self-Reflection in Academic Stress Assessment
Minyoung Park, **Bogyeom Park**, and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

A Self-Determination Theory-Based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement
Hyerim Han, **Bogyeom Park**, and Kyoungwon Seo
CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment
Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo
CHI EA '24: Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (33.9% acceptance; 391/1,154)

Multimodal Machine Learning Model for MCI Detection Using EEG, MRI and VR Data
Mariem Kallel, **Bogyeom Park**, Kyoungwon Seo, and Seong-Eun Kim
2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

Advancing Mild Cognitive Impairment Detection: Integrating VR, MRI, and Neuropsychological Insights for Comprehensive Diagnosis
Bogyeom Park, Jinseok Park, Hojin Choi, Hokyoung Ryu, and Kyoungwon Seo
2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

Early Screening of Mild Cognitive Impairment Using Multimodal VR-EP-EEG-MRI (VEEM) Biomarkers via Machine Learning
Se Young Kim, **Bogyeom Park**, Dohyun Kim, Hojin Choi, Jinseok Park, Hokyoung Ryu, and Kyoungwon Seo
2024 International Conference on Electronics, Information, and Communication (ICEIC), pp. 1-4

KOREAN CONFERENCE PAPERS

From Teacher Needs to Agentic AI: Designing and Validating a Personalized Career Counseling System
Bogyeom Park, Mina Yoo, Dongkuk Lee, Mi-ae Choi, Seona Park, So Young Jo, and Kyoungwon Seo
Proceedings of HCI Korea 2026 - Oral Presentation

The Impact of Self-Disclosing Chatbots for Academic Stress Assessment on Student Self-Reflection
Minyoung Park, **Bogyeom Park**, and Kyoungwon Seo
Proceedings of HCI Korea 2025, pp. 560-568 - Best Paper Award

Artificial Intelligence-Based Heritage Tree Disease Diagnosis Using Transfer Learning: A Case Study of Zelkova serrata
Sabin Lee, **Bogyeom Park**, Daejung Kim, and Kyoungwon Seo
Proceedings of HCI Korea 2024, pp. 212-219

Counterfactual vs. Prefactual: Two Narrative AIs Improve Causability for Health Data by Different Mechanisms
Hyobin Park (now **Bogyeom Park**) and Kyoungwon Seo
Proceedings of HCI Korea 2023, pp. 828-835

RESEARCH EXPERIENCE

Human-centered Artificial Intelligence (HAI) Lab, SeoulTech Seoul, South Korea
Graduate Researcher (Advisor: Kyoungwon Seo) *Mar. 2023 - Present*
Undergraduate Researcher *Jul. 2021 - Feb. 2023*
ICAP-Based AI Tutoring System for Probability and Statistics Learning
Research Assistant | Tutorus Labs *Mar. 2026 - Aug. 2026*
- Led the design and evaluation of an ICAP-based AI tutor that uses staged elicitation to promote active, constructive, and interactive engagement rather than answer delivery
- Developed an utterance-level coding and analytics workflow linking dialogue evidence with tutor correctness, usage patterns, and learning outcomes
- Built an interactive research dashboard for reviewing engagement labels and comparing AI-tutored learning processes

Agentic AI for Personalized Career, Academic, and Counseling Support
Research Assistant | Korea Education & Research Information Service *Jun. 2025 - Dec. 2025*
- Led a study of how agentic AI could support personalized educational pathways, counseling, and decision-making

AI Copilot Technologies for Adaptive, Teacher-Augmented Learning
Research Assistant | Institute for Information & Communications Technology Planning & Evaluation *Jul. 2023 - Dec. 2025*

- Built counseling chatbot and analysis models supporting AI-driven student coaching
- Co-developed an integrated platform enabling personalized teacher-augmented learning

AI-Based Early Screening and Prognosis Prediction for Landscape Tree Disease
Research Assistant | SUNGHA Co., Ltd. *Jul. 2023 - Dec. 2023*

- Built an expert-validated image dataset of Zelkova serrata, the species that accounts for more than half of Korea's legally protected trees
- Compared transfer-learning models with and without plant-disease pre-training for early screening of tree disease

VR-EP-EEG-MRI Digital Biomarker Basic Research Laboratory
Research Assistant | National Research Foundation of Korea *Mar. 2022 - Feb. 2024*

- Collected VR kiosk interaction data from participants with mild cognitive impairment and healthy controls in collaboration with Hanyang University Guri Hospital

Multimodal Deep Learning for Early Dementia Diagnosis from VR Daily-Living Data
Research Assistant | National Research Foundation of Korea *Mar. 2022 - Feb. 2024*

- Developed multimodal predictive models integrating VR behavioral and MRI biomarkers for early detection of mild cognitive impairment

TEACHING & MENTORING

2026 AX Academy Big Data Boot Camp *May 2026 - Aug 2026*
Tutor, Hyundai Motor Group

- Mentored nine participants from across the company, each developing an individual big data project from proposal to final deliverable
- Advised on analysis design and revised project plans with participants through the boot camp's project-based sprints

Deep Learning *Fall 2023*
Teaching Assistant, Seoul National University of Science and Technology

- Designed a final project using CNN-based models to predict drivers' physical and cognitive states from image data
- Supported lectures, advised student projects, and graded assignments and examinations

SKILLS

- **Research Methods -** experimental design, usability evaluation, user research, prototyping, multimodal learning analytics, statistical analysis, qualitative coding
- **AI and Data -** machine learning, deep learning, feature engineering, LLM applications, AI agents
- **Programming and Analysis -** Python, C/C#, SPSS
- **Design -** HCI theory, UX design, interaction prototyping, usability testing

PATENT

Explainable AI-Based System for Early Diagnosis and Prognosis Prediction of Alzheimer's Disease Using VR Biomarkers, and Method Thereof (Korean Patent Application No. 10-2023-0105821, filed Aug. 2023)

ACADEMIC SERVICE

- Student Volunteer, AIED 2026
- Publicity Manager, SeoulTech Human-centered Artificial Intelligence Lab